

Bayesian Inference

Lectures 3–4: Steps of Bayesian Analysis, Exchangeability, and the Beta–Bernoulli Model

Lecture 3

1 The Three Steps of Bayesian Data Analysis

Overall goal

Bayesian data analysis is about *finding the conditional distribution of unknown (parameter) quantities*, given the observed data.

Step 1. Setting up a full probability model: construct a joint probability distribution for *all* observable and unobservable quantities, consistent with the knowledge about the underlying scientific problem and the data-collection process.

Step 2. Conditioning on observed data: the posterior distribution is the conditional distribution of the unobserved quantities, given the observed data.

Step 3. Evaluating the fit of the model.

To derive / write up

For Step 1, write out explicitly what “joint distribution for all observable and unobservable quantities” means in a simple running example (e.g. coin tosses with unknown bias θ): what is observable ($y_1, \dots, y_n$), what is unobservable (θ), and what is the joint $p(\theta, y_1, \dots, y_n) = p(\theta) p(y_1, \dots, y_n | \theta)$? For Step 3, note (from your course) what specific diagnostics/checks are used to evaluate model fit (e.g. posterior predictive checks) once they are formally introduced.

2 Posterior Odds vs. Prior Odds

Setup: choosing between two competing values/hypotheses for the parameter, θ_1 or θ_2 (e.g. testing whether an estimator is unbiased, or comparing two models).

Definitions

$$\text{Prior odds: } \frac{p(\theta_1)}{p(\theta_2)}, \quad \text{Posterior odds: } \frac{p(\theta_1 | y)}{p(\theta_2 | y)}.$$

Derivation (Bayes’ rule applied to each hypothesis):

$$\frac{p(\theta_1 | y)}{p(\theta_2 | y)} = \frac{\frac{p(\theta_1) p(y | \theta_1)}{p(y)}}{\frac{p(\theta_2) p(y | \theta_2)}{p(y)}} = \frac{p(\theta_1) p(y | \theta_1)}{p(\theta_2) p(y | \theta_2)} = \underbrace{\frac{p(\theta_1)}{p(\theta_2)}}_{\text{prior odds}} \times \underbrace{\frac{p(y | \theta_1)}{p(y | \theta_2)}}_{\text{likelihood ratio (Bayes factor)}}.$$

Reading the comparison

$$\frac{p(\theta_1 | y)}{p(\theta_2 | y)} \geq \frac{p(\theta_1)}{p(\theta_2)} \iff \frac{p(y | \theta_1)}{p(y | \theta_2)} \geq 1,$$

i.e. the posterior odds move away from the prior odds in the direction favored by the likelihood ratio: if the data are more likely under θ_1 than θ_2 , the posterior odds shift toward θ_1 relative to the prior odds, and vice versa.

As $n \rightarrow \infty$

The notes flag the behavior of $\frac{p(\theta_1 | y)}{p(\theta_2 | y)}$ as the sample size n grows (written schematically as $n, n, n, n, \dots \rightarrow$). With more and more data, the likelihood ratio should dominate the fixed prior odds, and the posterior odds should concentrate on whichever of θ_1, θ_2 is (closer to) the truth.

To derive / write up

State and prove this consistency result precisely: under repeated iid sampling, show that $p(y_1, \dots, y_n | \theta_1) / p(y_1, \dots, y_n | \theta_2) \rightarrow 0$ or ∞ a.s. depending on which hypothesis generated the data (a Kullback–Leibler / law-of-large-numbers argument on the log-likelihood ratio).

3 Conditional Independence and Exchangeability

Motivating example (from the notes): a sequence of observations $y_1, \dots, y_{10}$ from 10 people, with an 11th person's observation y_{11} to be predicted (e.g. measurements in the range 100–200, such as heights in cm).

Conditional independence given θ

Given the parameter θ , a new observation is independent of the past observations:

$$p(y_{n+1} | \theta, y_1, \dots, y_n) = p(y_{n+1} | \theta).$$

Marginally (i.e. before conditioning on θ), however,

$$p(y_{n+1} | y_1, \dots, y_n) = f(y_1, \dots, y_n)$$

is *not* simply $p(y_{n+1})$ — the past data inform beliefs about θ , which in turn inform the prediction of y_{n+1} . This is the basic mechanism of *learning from data* in the Bayesian framework.

Definition: Exchangeability

$y_1, \dots, y_n$ are **exchangeable** if their joint distribution $p(y_1, \dots, y_n)$ is *invariant with respect to permutation of the indices*: for any permutation π of $\{1, \dots, n\}$,

$$p(y_1, \dots, y_n) = p(y_{\pi(1)}, \dots, y_{\pi(n)}).$$

In particular, for a new observation,

$$p(y_{n+1} | y_1, \dots, y_n) = p(y_{n+1} | y_2, y_2, y_1, y_1, \dots, y_n)$$

(the specific labeling/ordering of the conditioning variables does not matter).

Why this matters

Exchangeability is a weaker, more natural assumption than “iid” to justify from modeling considerations directly, yet (via de Finetti’s representation theorem, Lecture 4) it turns out to be *equivalent* to the existence of a parameter θ and a prior $p(\theta)$ such that, conditional on θ , the y_i are iid. This is the conceptual bridge between “exchangeable data” and “iid-given- θ data with a prior on θ .”

To derive / write up

Write out carefully why $\text{iid} \Rightarrow \text{exchangeable}$ but not conversely in general (an exchangeable sequence need not be iid *unconditionally* — only conditionally on θ under de Finetti’s theorem). A short two-observation counterexample (e.g. draws without replacement from a finite population) is a good self-check.

Lecture 4

4 De Finetti's Representation Theorem

Setup: binary sequences, e.g. $1, 0, 0, 1, 0, \dots$ or $1, 1, 0, 0, 0, \dots$, with $\sum_{i=1}^n Y_i$ as the natural summary statistic.

De Finetti representation

If $y_1, y_2, \dots$ is an (infinitely extendable) exchangeable sequence of binary outcomes, then there exists a distribution $p(\theta)$ such that

$$p(y_1, \dots, y_n) = \int_0^1 \theta^{\sum_i y_i} (1 - \theta)^{n - \sum_i y_i} p(\theta) d\theta.$$

Equivalently: $Y_1, \dots, Y_n \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta)$ for some random $\theta \sim p(\theta)$, and

$$\sum_{i=1}^n Y_i \mid \theta \sim \text{Binomial}(n, \theta).$$

To derive / write up

This is the theorem that formally justifies “acting as if” the data are iid given a parameter θ , whenever all you are willing to assume is exchangeability. State the theorem's hypotheses precisely (infinite exchangeability, not just finite n) and note why finite exchangeability alone is *not* sufficient for the representation to hold exactly.

5 The Beta–Bernoulli Model

Model:

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta), \quad \mathbf{y} = (y_1, \dots, y_n).$$

Prior – likelihood – posterior

$$\underbrace{p(\theta)}_{\text{prior}}, \quad \underbrace{p(\mathbf{y} \mid \theta)}_{\text{likelihood}}, \quad \underbrace{p(\theta \mid \mathbf{y})}_{\text{posterior}} \propto p(\theta) p(\mathbf{y} \mid \theta).$$

Beta density: for $X \sim \text{Beta}(a, b)$,

$$f_X(x) \propto x^{a-1} (1-x)^{b-1}, \quad 0 < x < 1.$$

Posterior derivation

With likelihood $p(\mathbf{y} \mid \theta) = \theta^{\sum y_i} (1 - \theta)^{n - \sum y_i}$, and writing $z = \sum_i y_i$,

$$p(\theta \mid \mathbf{y}) \propto p(\theta) \theta^z (1 - \theta)^{n-z}.$$

4.1 The Uniform Prior Special Case

Take $\theta \sim U(0, 1)$, i.e. $p(\theta) = 1$ on $(0, 1)$. Then

$$p(\theta \mid \mathbf{y}) \propto \theta^z (1 - \theta)^{n-z} \implies \theta \mid \mathbf{y} \sim \text{Beta}(z + 1, n - z + 1).$$

Marginal distribution of z under the uniform prior

With $p(\theta) = 1$, the marginal (prior predictive) distribution of $z = \sum_i Y_i$ is *discrete uniform* on $\{0, 1, \dots, n\}$:

$$z \in \{0, 1, \dots, n\}, \quad \text{each with probability } \frac{1}{n+1}.$$

To derive / write up

Derive this marginal result directly:

$$p(z) = \binom{n}{z} \int_0^1 \theta^z (1-\theta)^{n-z} d\theta,$$

evaluate the integral using the Beta function $B(z+1, n-z+1) = \frac{z!(n-z)!}{(n+1)!}$, and simplify to confirm $p(z) = \frac{1}{n+1}$ for every $z \in \{0, \dots, n\}$. This is a classical (and slightly surprising) special property of the uniform prior on a Bernoulli success probability.

4.2 Posterior Prediction

Goal: predict a new observation $\tilde{y}$ given the data $\mathbf{y}$.

Posterior predictive probability

$$\Pr(\tilde{y} = 1 \mid \mathbf{y}) = \int_0^1 \Pr(\tilde{y} = 1 \mid \theta, \mathbf{y}) p(\theta \mid \mathbf{y}) d\theta = \int_0^1 \theta p(\theta \mid \mathbf{y}) d\theta = \mathbb{E}[\theta \mid \mathbf{y}].$$

Using $\theta \mid \mathbf{y} \sim \text{Beta}(z+1, n-z+1)$ and the Beta mean formula $\mathbb{E}(X) = \frac{a}{a+b}$ for $X \sim \text{Beta}(a, b)$:

$$\Pr(\tilde{y} = 1 \mid \mathbf{y}) = \mathbb{E}[\theta \mid \mathbf{y}] = \frac{z+1}{(z+1) + (n-z+1)} = \frac{z+1}{n+2}.$$

Laplace's rule of succession

$\frac{z+1}{n+2}$ is the classical “rule of succession”: compare with the maximum likelihood estimate $\hat{\theta}_{\text{MLE}} = z/n$. The posterior mean shrinks the raw proportion slightly toward $1/2$, with the effect vanishing as $n \rightarrow \infty$.

To derive / write up

Verify $\frac{z}{n}$ vs. $\frac{z+1}{n+2}$ numerically for a few small (z, n) and confirm they converge as n grows. Also derive $\Pr(\tilde{y} = 1 \mid \mathbf{y}) = \mathbb{E}[\theta \mid \mathbf{y}]$ generally (not just for this conjugate case) by justifying the swap of integration order $\int p(\tilde{y} \mid \theta) p(\theta \mid \mathbf{y}) d\theta$.

4.3 Summarizing the Posterior

For $\theta \mid \mathbf{y} \sim \text{Beta}(z+1, n-z+1)$, the standard summaries are:

Quantity	Notation
Posterior mean	$\mathbb{E}(\theta \mid \mathbf{y}) = \frac{z+1}{n+2}$
Posterior median	$\text{Med}(\theta \mid \mathbf{y})$
Posterior mode	$\text{Mode}(\theta \mid \mathbf{y})$
Posterior standard deviation	$\text{SD}(\theta \mid \mathbf{y})$
Maximum likelihood estimate	$\hat{\theta}_{\text{MLE}} = z/n$

The posterior density $\text{Beta}(z+1, n-z+1)$ is unimodal on $(0, 1)$, rising from 0 at $\theta = 0$ to a peak and back down to 0 at $\theta = 1$ (sketch: a hump-shaped curve on $[0, 1]$, skewed depending on z relative to n).

Credible interval percentiles noted in the margin

The values 0.05 and 0.95 are flagged — these are the standard lower/upper quantiles used to construct a (two-sided) 90% posterior credible interval for θ :

$$[\theta_{0.025}, \theta_{0.975}] \text{ or } [\theta_{0.05}, \theta_{0.95}],$$

depending on the convention (95% vs. 90%) being used — confirm which one your course defines.

To derive / write up

- Derive the mode of $\text{Beta}(a, b)$ for $a, b > 1$: $\text{Mode} = \frac{a-1}{a+b-2}$, and specialize to $a = z+1, b = n-z+1$.
- Derive the variance of $\text{Beta}(a, b)$: $\text{Var}(X) = \frac{ab}{(a+b)^2(a+b+1)}$, specialize similarly, and take the square root for $\text{SD}(\theta \mid \mathbf{y})$.
- Compare $\hat{\theta}_{\text{MLE}} = z/n$, the posterior mean $(z+1)/(n+2)$, and the posterior mode z/n (note: under a uniform prior the posterior *mode* coincides with the MLE, while the posterior *mean* does not) — confirm this coincidence algebraically.
- Numerically integrate (or use the incomplete Beta function) to compute the 0.05 and 0.95 quantiles of $\text{Beta}(z+1, n-z+1)$ for a specific (z, n) from class, to build intuition for how the credible interval narrows as n grows.